

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

~~Third~~ **Fourth** Semester of B. Tech. Examination (CL)
November – December 2011

CL 208 Advanced Surveying

Date: 06.12.2011, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain tangential method of tacheometry surveying [05]
(b) Enlist key components of GIS and explain any one. [02]
- Q - 2 (a) What are the common errors and mistakes encountered in tacheometry? [06]
(b) The following are observed values of angle with a theodolite [06]

Angle	20° 20' 20"	20° 20' 00"	20° 20' 40"
Weight	2	2	3

Find probable error in 1) single observation of unit weight 2) weighted arithmetic mean.

- (c) What is an analytic lens? Why it is used? [02]

OR

- Q - 2 (a) Determine the distance QR and elevation of station Q from following observations made with a tacheometer fitted with analytic lens. Take RL of BM = 250.50 m. [06]

Inst. Station	Staff. Station	Vertical Angle	Staff Readings		
R	BM	-5° 30'	1.255	1.810	2.365
R	Q	+10° 6'	1.30	2.120	2.940

- (b) Form the normal equation of x, y and z in the following equations of equal weight : [06]

$$3x + 3y + z - 4 = 0, \quad x + 2y + 2z - 6 = 0, \quad 5x + y + 4z - 21 = 0$$

If weights of above equations are 2, 3 and 1 respectively form normal equation of x, y and z.

- (c) What is subtense bar? [02]

- Q - 3 (a) Write a short note on GPS. [06]

- (b) What are the limitations of remote sensing? [02]

- (c) State and prove principle of least square [06]

OR

- Q - 3 (a) Explain in brief 'visual image interpretation'? [06]
(b) Write briefly about applications of GIS. [02]
(c) State laws of weight. [06]

SECTION - II

- Q - 4 (a) Explain the basic principle of trigonometric leveling. [03]
(b) Differentiate between Terrestrial and Aerial photogrammetry [04]
Q - 5 (a) Explain different triangulation systems with the help of neat sketch. [04]
(b) What are the different types of signals used for triangulation? [04]
(c) Explain the following terms in connection with aerial survey : [06]

Oblique Photograph, Ground Nadir Point, Principal Point, Principal Plane, Principal Line, Perspective Projection

OR

- Q - 5 (a) Enlist various methods of base line measurement and explain any one of them. [04]
(b) What do you mean by a base line? Explain how would you prolong a base line? [04]
(c) Define scale of a photograph and explain datum scale. [06]

A vertical photograph is taken from an altitude 1000m above mean seal level. Determine scale of the photograph for terrain at elevation 50m and 100m if focal length of camera is 15cm.

- Q - 6 (a) An instrument was set up at station P and the angle of elevation to a vane 3m above foot of staff at station Q was observed as 10° . If the horizontal distance between P and Q is 1500m determine the R.L. of station Q with the help of given R.L. of P as 2500m. [04]
(b) 1. Draw neat sketch showing the components of an aerial camera. [05]
2. Explain the terms Declination and Azimuth
(c) The standard time meridian in India is $82^{\circ} 30' E$. If the standard time at any instant is 18 hr. 14 min 10 sec, find the local mean time for place having longitudes 1) $20^{\circ} E$ 2) $20^{\circ} W$. [05]

OR

- Q - 6 (a) Describe the procedure of measuring elevation of a station with trigonometric levelling when distance to the station cannot be measured. [04]
(b) 1. Explain Photomaps. [05]
2. Explain Ecliptic with neat sketch
(c) Write a brief note on Solar apparent time and Standard time [05]
